

Yash Mahalwal

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Summary

Senior Software Engineer with 5+ years of experience building and scaling high performance web and mobile products used by over 10 million users. Strong focus on frontend development using React, Next.js, and TypeScript, with hands on experience building backend systems using AWS, GraphQL, and event driven architectures. Proven track record of improving performance, conversion, and system scalability, including reducing latency by over 80 percent and increasing conversion by 30 percent. Experience building agent based workflows and developer tooling.

Experience

Senior Software Engineer, GrowthDay – Remote Dec 2023 – April 2026

- Redesigned payment flow from synchronous processing to an event driven pipeline with queued execution and idempotent endpoints, reducing latency from 12–17s to under 3s and improving conversion.
- Scaled real-time system from 500 to 100K+ interactive participants and up to 10M viewers for large live events by replacing peer-to-peer architecture with SFU based architecture and HLS streaming, reworking interaction models to avoid N² event patterns and maintain performance at scale.
- Reduced LCP from 8s to under 1.5s by migrating to Next.js SSR and React Server Components, leveraging streaming, caching, and static generation, and eliminating client-side waterfalls.
- Drove product and growth initiatives through A/B testing and behavioral analytics, partnering with product and marketing to improve engagement, retention, and conversion, while mentoring engineers on experimentation and data-driven development.
- Architected high throughput, reliable distributed backend workflows for user onboarding, migration, and external integrations using Lambda, AppSync, DynamoDB, Step Functions, and SQS, with robust retry and failure handling, increasing automation efficiency by 80%.
- Contributed to mobile application development (Kotlin Compose and SwiftUI), building and shipping product features across platforms.

Frontend Developer, GrowthDay – Remote July 2021 – Dec 2023

- Built a real-time discussion platform using WebSockets and WebRTC (peer-to-peer), enabling video, chat, and live interactions across web and mobile.
- Improved OTT video experience by introducing adaptive streaming, better buffering, and a persistent mini player, increasing video completion rates by 35%.
- Built a shared cross-platform module with automated synchronization via CI, unifying logic, tests, and types across web and mobile, reducing duplication by 50%.
- Refactored poorly performing web and mobile codebases, optimizing bundle size, network usage, and rendering, reducing infrastructure costs by 50% and preventing server overload under high traffic.
- Led delivery of 20+ features across cross-functional pods, driving architecture decisions, coordinating with product, and unblocking teams, reducing rollout time by 40%.

Product Analyst Intern, American Express – Gurugram, India (Remote) Jan 2021 – June 2021

- Built an error handling and management system for machine learning model deployments, reducing unhandled failures by 70%.
- Enabled deployment of new machine learning models through internal pipelines, streamlining model onboarding and production rollout.
- Developed distributed training solutions using Scala and Hadoop for internal machine learning platforms.

Product Developer Intern, Appointy Software Inc – Bhopal, India (On-site) May 2019 – Apr 2020

- Designed a scalable routing system with nested layouts and dynamic routing, improving flexibility and maintainability of the frontend architecture.

- Built a UI recurrence rule engine for appointment scheduling, increasing scheduling efficiency by 3x.
- Developed an intuitive drag-and-drop form builder, reducing onboarding time for business clients by 60%.
- Implemented SSR with GraphQL, improving SEO and increasing organic traffic by 35%.
- Worked with Golang gRPC microservices exposed via GraphQL APIs, contributing to scalable backend infrastructure and seamless data flow.

Education

Maulana Azad National Institute of Technology, B.Tech in Computer Science

July 2017 – July 2021

- CGPA: 8.93/10

Projects

Fark.ai: Agentic GitHub workflow that analyzes backend pull requests for breaking API changes and cross-validates against frontend codebases to detect real-world impact, improving developer feedback and reducing integration issues.

Parcel Plugin Add Source: Parcel transformer to add source code to assets, handy for documentation.

React Synced State: React hook to control and synchronize state across React and React Native apps.

Publications

Optimising React Apps: Walkthrough of React performance challenges, tradeoffs, and mitigation strategies.

Sharing code between projects: Comparison of strategies for multi-project code reuse with pros/cons.

The Heisenberg Manoeuvre: Exploration of how npm packages can be misused to exploit dependency chains, demonstrating risks and implications for developers.

Relay modern: SSR and Next.js: Approaches to integrating Relay GraphQL with server-side rendering in Next.js, including tradeoffs and implementation patterns.

Technologies

Frontend & Mobile: TypeScript, React, React Native, Next.js, React Server Components, WebRTC, Kotlin Compose, SwiftUI, Zustand, Redux Toolkit, Storybook, Webpack, Vite, Material-UI, Chakra UI, Ant Design (AntD), Tailwind CSS

Backend & Cloud: Node.js, WebSockets, REST, GraphQL, AWS (Cognito, DynamoDB, AppSync, S3, EventBridge, Lambda, CDK), Golang, Python, MySQL, gRPC, Protocol Buffers, Docker, Kubernetes, CI/CD

Testing & Performance: Jest, Cypress, Playwright, Vitest, A/B Testing, Lighthouse, Web Vitals, Performance Profiling & Optimization

AI & Tooling: LLM-based agents, OpenAI SDK, Vercel AI SDK, agent workflows, AI-assisted development (Cursor, Codex, Claude)